CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 18: Tangents and Intersecting Chords

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the length of the tangent drawn to a circle of radius 9 cm from a point 15 cm from its centre.

2. The tangent from a point P to a circle is 24 cm long and the radius is 7 cm. Find the distance of P from the centre.

3. PA and PB are tangents to a circle with centre O. If $\angle AOB = 124^\circ$, find $\angle APB$ and $\angle OPA$.

4. Two circles with radii 9 cm and 4 cm touch externally. Find the distance between their centres, and the length of a direct common tangent.

5. Chords AB and CD of a circle intersect at P inside it. If $AP = 5$ cm, $PB = 6$ cm and $CP = 3$ cm, find PD.

6. Chords AB and CD of a circle intersect inside it at P. If $AB = 11$ cm, $PB = 3$ cm and $PD = 4$ cm, find PC.

7. From an external point P, a secant PAB and a tangent PT are drawn. If $PA = 3$ cm and $PB = 12$ cm, find PT.

8. A tangent PT is 8 cm long and a secant from P cuts the circle at A and B with $PA = 4$ cm. Find AB.

9. The tangent at A to a circle makes an angle of 62° with a chord AB. Find the angle that AB subtends at a point in the alternate segment, and at the centre.

10. A circle touches the sides of a quadrilateral $ABCD$. If $AB = 9$ cm, $BC = 7$ cm and $CD = 5$ cm, find AD.

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Answer Key

1. 12 cm
2. 25 cm
3. 56° and 28°
4. 13 cm; $\sqrt{13^2 - 5^2} = 12$ cm
5. 10 cm
6. $8 \times 3 = PC \times 4$, so $PC = 6$ cm
7. 6 cm
8. $PB = 16$ cm; $AB = 12$ cm
9. 62° and 124°
10. 7 cm

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